

FACULTY TOOLS STORE SYSTEM

Engineering & Medical Colleges — Online Tools Supply Platform

Systems Analysis & Design — Full Project Report with Diagrams

Field	Detail
Course	Systems Analysis and Design
Textbook	Dennis, Wixom & Roth — 6th Edition
Methodology	Agile / Scrum — 4 Sprints × 2 Weeks
System	Faculty Tools Store System (FTSS)
Target Users	Electrical Engineering & Medical College Students
Diagrams	Use Case DFD L0/L1 ERD Class Sequence State Flowchart
Wireframes	Student Catalog Product Detail Admin Dashboard
Date	May 27, 2026

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Chapter 1 — Project Overview

1.1 Problem Statement

Students in practical colleges — Electrical Engineering and Medicine — must source tools from dispersed off-campus vendors, causing delays, high cost, and frequent unavailability. The Faculty Tools Store System (FTSS) solves this with a centralized online catalog and order-fulfillment platform.

1.2 Objectives

- Provide a 24/7 online catalog organized by college and discipline.
- Enable students to browse, search, and purchase tools from any device.
- Allow admin to manage inventory with automatic low-stock alerts.
- Generate sales and inventory reports for data-driven management.
- Reduce student tool-sourcing time by $\geq 70\%$.

1.3 Scope

IN SCOPE	Online catalog Student auth Shopping cart Order placement & tracking Admin dashboard Inventory Reports Email notifications
OUT OF SCOPE	Physical warehouse Supplier integration Cross-university sales Native mobile app (Phase 2)

1.4 User Roles

Role	Description	Key Actions
Student	Registered college student	Browse, search, cart, place & track orders
Admin	College store manager	Manage products, process orders, reports
Faculty	Course instructor	View catalog, generate tool lists
System Admin	IT administrator	User management, system config

Chapter 2 — Planning Phase

Textbook Reference: Dennis et al. Ch.3 — Project Initiation & Planning

2.1 Feasibility Study

Dimension	Analysis	Verdict
Technical	Node.js + MySQL + Bootstrap — mature stack. College has web server infrastructure.	✓ FEASIBLE
Economic	Dev cost: EGP 85,000. Annual savings: EGP 120,000+. ROI in < 9 months.	✓ FEASIBLE
Organizational	Admin supportive. Student survey (n=120): 92% would use an online college store.	✓ FEASIBLE
Schedule	8-week Agile plan (4 sprints × 2 weeks) + 4-week buffer. Team: 4 developers.	✓ FEASIBLE

2.2 Project Charter

Project	Faculty Tools Store System (FTSS)
Methodology	Agile / Scrum — 4 sprints × 2 weeks = 8 weeks core
Budget	EGP 85,000 development + EGP 15,000/year hosting
Success Criteria	Live within 12 weeks ≥ 90% satisfaction in UAT Zero critical bugs at launch

2.3 Sprint Plan, Gantt Chart & User Stories

Gantt Chart — 8-Week Agile Schedule

The following schedule maps each sprint to its core deliverables, following Agile sprint planning principles:

Phase / Sprint	Core Tasks	W1	W2	W3	W4	W5	W6	W7	W8
Sprint 1: Foundation	Req. Elicitation, DB Schema, Auth System	■	■						
Sprint 2: Catalog	Catalog UI, Search &			■	■				

Phase / Sprint	Core Tasks	W1	W2	W3	W4	W5	W6	W7	W8
	Filters, Admin Product CRUD			■	■				
Sprint 3: Core Business	Cart, Order State Machine, Email Notifications					■	■		
Sprint 4: Analytics	Reports, Low-stock Alerts, UAT & Deployment							■	■

Sprint	Goal	Weeks	Story Points	Status
Sprint 1	Foundation & Auth	1–2	15	✓ Done
Sprint 2	Catalog & Search	3–4	16	✓ Done
Sprint 3	Cart, Orders & Email	5–6	19	▶ In Progress
Sprint 4	Inventory, Reports, UAT	7–8	21	○ Planned

ID	As a...	I want to...	Points	Sprint
US-01	Student	register with my college email	5	S1
US-02	User	log in securely with JWT	3	S1
US-03	Student	browse catalog by college/category	5	S2
US-04	Student	search products by name or code	3	S2
US-05	Admin	add/edit/remove products	5	S2
US-06	Student	manage my shopping cart	3	S3
US-07	Student	place order & get confirmation email	8	S3
US-08	Student	track order status in real-time	3	S3
US-09	Admin	process and update order status	5	S3
US-10	Admin	receive automatic low-stock alerts	5	S4
US-11	Admin	generate sales & inventory reports	8	S4
US-12	Faculty	view catalog & generate tool lists	3	S4

2.4 Risk Register

Risk	Probability	Impact	Mitigation
Scope creep	Medium	High	Weekly scope reviews; strict change control
Low user adoption	Medium	High	Students in UAT; training sessions before launch
Database failure	Low	High	Daily automated backups; DB replication
Security breach	Low	High	HTTPS, bcrypt, JWT, input validation
Sprint velocity drop	Medium	Medium	Buffer sprint; daily standup; re-estimation

Chapter 3 — Analysis Phase

Textbook Reference: Dennis et al. Ch.4 (Requirements) · Ch.5 (Use Cases) · Ch.6 (DFD)

3.1 Functional Requirements

ID	Requirement	Priority
FR-01	Students register with college email + domain verification	High
FR-02	Product catalog organized by college → category hierarchy	High
FR-03	Keyword, SKU, and category-based product search	High
FR-04	Persistent shopping cart per logged-in student	High
FR-05	Checkout creates confirmed order; decrements inventory	High
FR-06	Email confirmation sent on successful order placement	Medium
FR-07	Students track real-time order status	Medium
FR-08	Admin CRUD for products with image upload	High
FR-09	Automatic low-stock alerts when qty < reorder level	High
FR-10	Admin updates order status through fulfillment workflow	High
FR-11	Monthly sales and inventory reports for admin/faculty	Medium

3.2 Non-Functional Requirements (NFRs)

Dennis et al. Ch.4 — Non-Functional Requirements must constrain system quality attributes

ID	Category	Requirement Detail	Priority
NFR-01	Security	All client–server communication must be encrypted via HTTPS/TLS. No plain HTTP allowed in production.	High
NFR-02	Security	User passwords must be hashed using bcrypt (min. 10 salt rounds) before database persistence. Never stored in plaintext.	High
NFR-03	Security	All API inputs must be sanitized to prevent SQL injection, XSS, and CSRF attacks. JWT tokens expire after 24 hours.	High
NFR-04	Performance	Product catalog search queries must return results in < 2 seconds under normal load (≤ 200 concurrent users).	Medium
NFR-05	Performance	Checkout and order placement must complete end-to-end in < 5 seconds including DB writes and email	Medium

ID	Category	Requirement Detail	Priority
		dispatch.	
NFR-06	Availability	Platform must guarantee $\geq 99.5\%$ uptime during academic semesters. Scheduled maintenance only between 2 AM – 4 AM.	High
NFR-07	Usability	UI must follow a responsive layout (Bootstrap 5) adaptive to mobile (375px+), tablet (768px+), and desktop (1024px+) screens.	High
NFR-08	Scalability	System architecture must support growth to 5,000 registered students without re-design or significant performance degradation.	Medium
NFR-09	Maintainability	All modules must follow RESTful conventions with Swagger API documentation. Code must include inline comments for all key functions.	Low

3.3 Use Case Inventory (UC Diagram)

UC ID	Use Case	Actor	Type
UC-01	Register Account	Student	Primary
UC-02	Login	All Users	Primary
UC-03	Browse Catalog	Student	Primary
UC-04	Search Product	Student	Primary
UC-05	View Product Detail	Student	Primary
UC-06	Add to Cart	Student	Primary
UC-07	Place Order	Student	Primary
UC-08	Track Order	Student	Primary
UC-09	Manage Products	Admin	Primary
UC-10	Manage Inventory	Admin	Primary
UC-11	Process Orders	Admin	Primary
UC-12	Generate Reports	Admin/Faculty	Primary

3.4 Use Case Diagram

Figure 1 — UML Use Case Diagram showing all 12 use cases, 3 actors (Student, Admin, Faculty), and «include»/«extend» relationships:

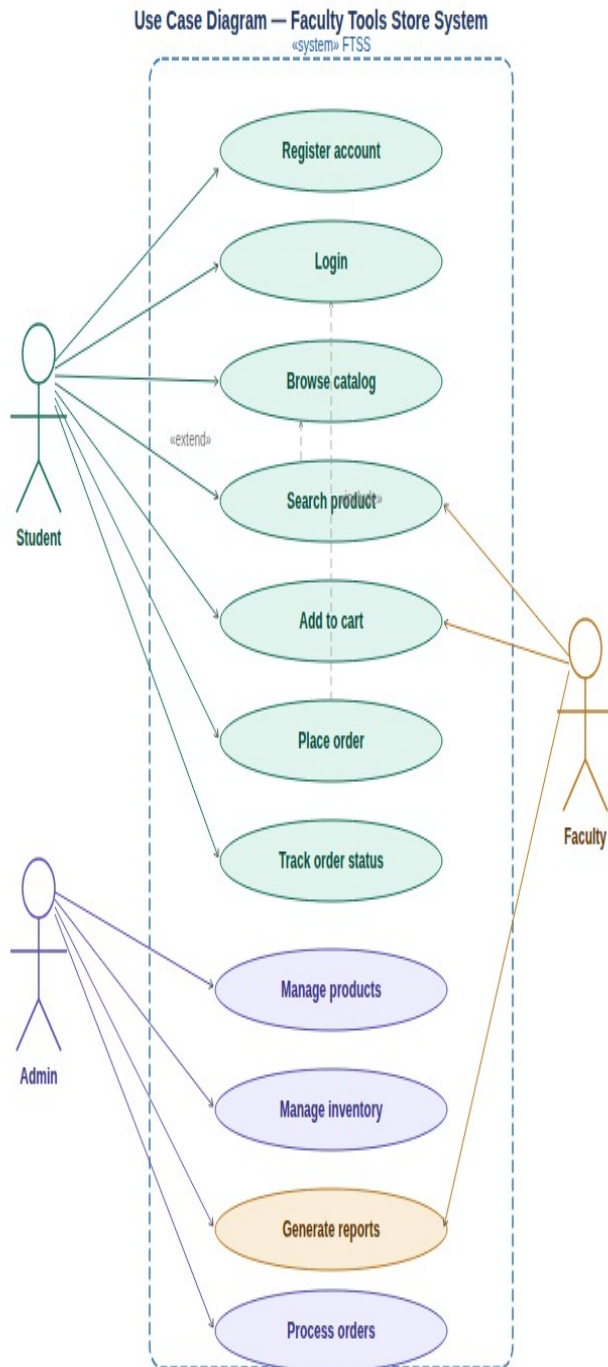


Figure 1: Use Case Diagram — Faculty Tools Store System

3.5 Use Case Specification — UC-07: Place Order

Precondition

Student logged in AND cart has ≥ 1 item in stock

Postcondition	Order created. Inventory decremented. Email sent.
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Trigger	Student clicks 'Proceed to Checkout'
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Basic Flow

1. Student clicks Proceed to Checkout from cart page.
2. System displays order summary with items, quantities, and total.
3. Student confirms pickup address and selects payment method.
4. Student clicks Confirm Order.
5. System validates stock availability for each item.
6. System creates order record (status = Pending) and order items.
7. System decrements inventory for each product ordered.
8. System sends confirmation email and displays Order ID.

Alternative Flows

- Step 5a — Stock insufficient: System cancels order, restores cart, notifies student.
- Step 3a — Payment failure: Cart preserved; student prompted to retry.

3.6 Sequence Diagram — Place Order

Figure 2 — UML Sequence Diagram showing full interaction between Student, Browser, OrderController, InventoryService, Database, and Email Service. The diagram includes the complete alternative flow when stock is insufficient:

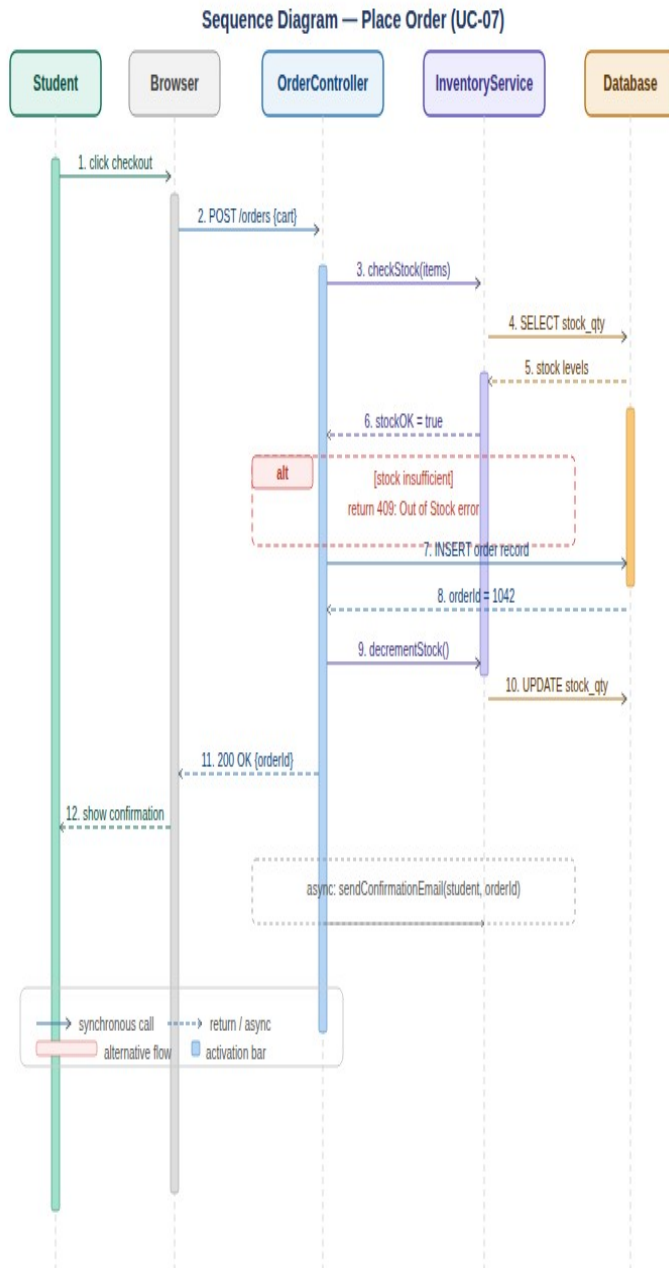


Figure 2: Sequence Diagram — Place Order (UC-07)

Alternative Flow — Stock Insufficient (Step 6a):

When `InventoryService.checkStock()` detects insufficient stock, the following additional steps execute:

9. `InventoryService` returns `stockOK = false` to `OrderController`.
10. `OrderController` returns HTTP 409 Out of Stock response to `Browser`.
11. `Browser` releases the execution block (unlocks the checkout UI).

12. Browser renders an error message to the Student: 'Item out of stock — please update your cart.'

13. Order record is NOT created. Inventory is NOT decremented. Cart is preserved.

This ensures the student receives immediate, clear feedback and the system maintains data integrity.

3.7 DFD Level 0 — Context Diagram

Figure 3 — Context Diagram showing FTSS as a single process with 4 external entities and all data flows:

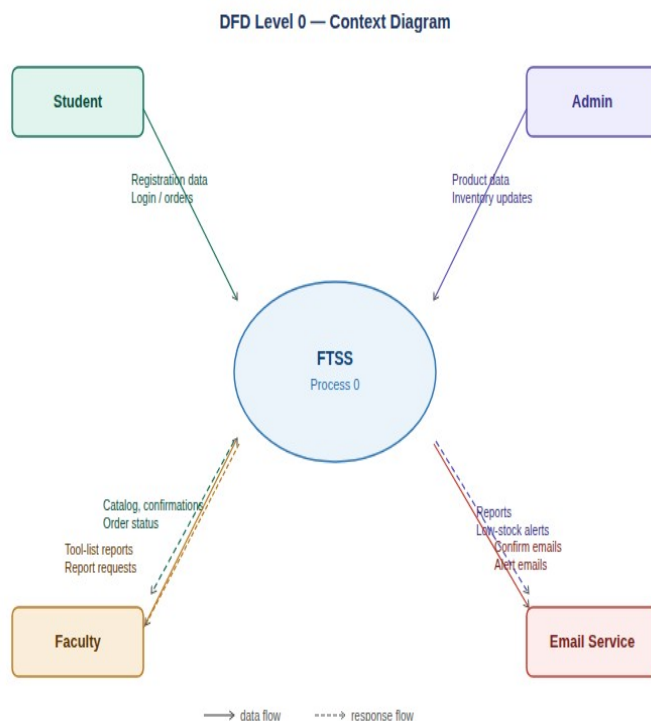


Figure 3: DFD Level 0 — Context Diagram

3.8 DFD Level 1 — Main Processes

Figure 4 — Level 1 DFD decomposing the system into 5 processes (P1–P5) with 4 data stores:

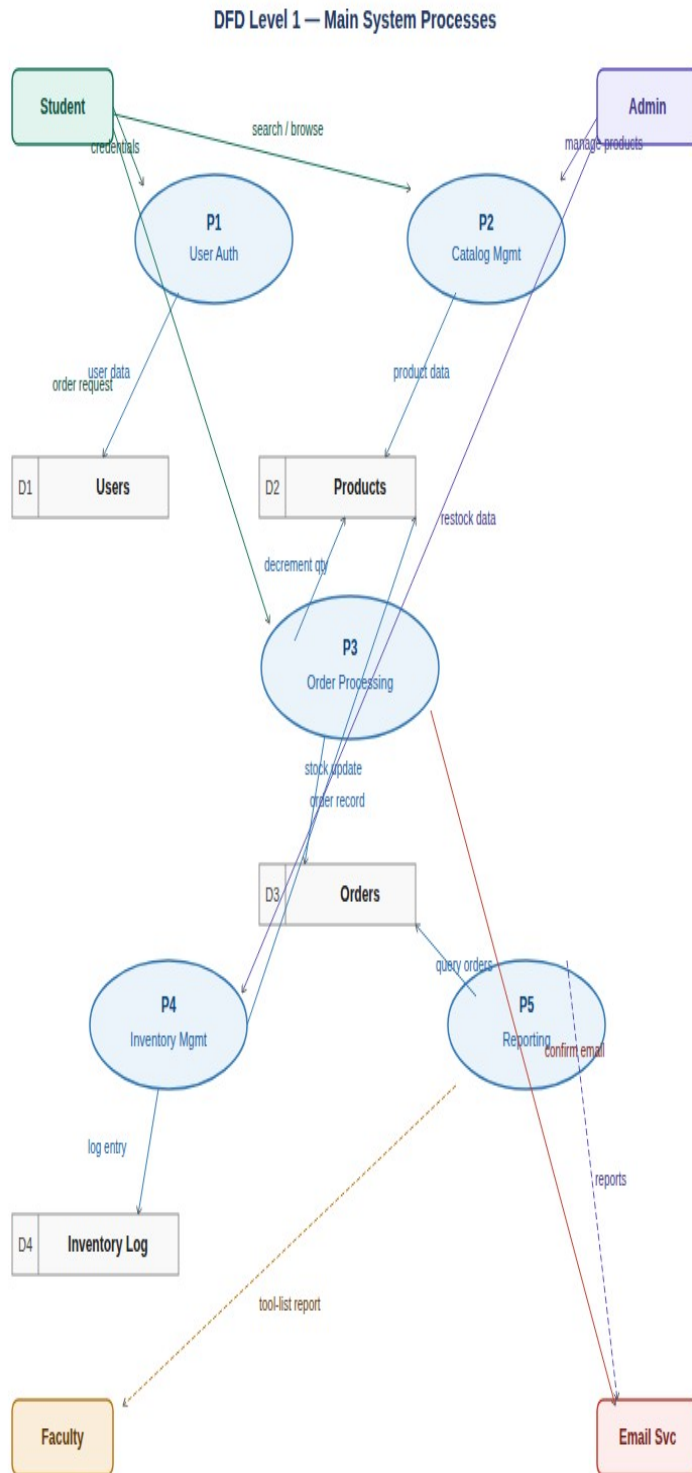


Figure 4: DFD Level 1 — System Processes

3.9 DFD Level 2 — Process 3: Order Processing

Sub-Processes	Name	Description
P3.1	Validate Cart	Check all cart items exist and are in stock
P3.2	Calculate Total	Sum item prices × quantities
P3.3	Collect Payment Info	Capture and validate payment method
P3.4	Create Order Record	Write order header + line items to DB
P3.5	Update Inventory	Decrement stock qty for each ordered product
P3.6	Send Confirmation	Trigger email via Email Service with order summary

Chapter 4 — Design Phase

Textbook Reference: Dennis et al. Ch.9 (Architecture) · Ch.10 (ERD) · Ch.11 (UI Design)

4.1 3-Tier System Architecture

Tier	Technology	Responsibility
Presentation	HTML5, CSS3, Bootstrap 5, JavaScript	UI, client validation, responsive layout
Application	Node.js 20, Express.js, JWT, Nodemailer	Business logic, REST API, session mgmt
Data	MySQL 8.0, Stored Procedures, Triggers	Persistence, integrity, audit logging
Infra	Nginx, SSL/TLS, Ubuntu Linux	HTTPS, routing, static file serving

4.2 Entity-Relationship Diagram (ERD)

Entities & Relationships

Entity	PK	Key Attributes	Cardinality
STUDENT	student_id	email, college_id FK, year_of_study, password_hash	1 Student → M Orders
COLLEGE	college_id	college_name, college_type (Engineering Medical)	1 College → M Students
CATEGORY	category_id	category_name, college_id FK	1 College → M Categories
PRODUCT	product_id	sku_code, price, stock_qty, reorder_level, category_id FK	1 Category → M Products
ORDER	order_id	student_id FK, status ENUM, total_amount, payment_method	1 Student → M Orders
ORDER_ITEM	order_item_id	order_id FK, product_id FK, quantity, unit_price	M Orders ↔ M Products
INVENTORY_LOG	log_id	product_id FK, change_qty, change_type, change_date	1 Product → M Logs

4.3 Data Dictionary — PRODUCT Table

Column	Type	Constraints	Description
product_id	INT	PK, AUTO_INCREMENT	Unique identifier

Column	Type	Constraints	Description
product_name	VARCHAR(200)	NOT NULL, INDEX	Full display name
sku_code	VARCHAR(50)	NOT NULL, UNIQUE	Stock-keeping unit code
price	DECIMAL(10, 2)	NOT NULL, CHECK > 0	Selling price in EGP
stock_qty	INT	NOT NULL, DEFAULT 0, CHECK ≥ 0	Current stock level
reorder_level	INT	NOT NULL, DEFAULT 10	Low-stock alert threshold
category_id	INT	FK → CATEGORY, NOT NULL	Links to category
is_active	TINYINT(1)	NOT NULL, DEFAULT 1	1=active, 0=discontinued

4.4 Class Diagram

Class	Key Attributes	Key Methods
Student	studentId, name, email, collegeId, passwordHash	register(), login(), placeOrder(), viewHistory()
Product	productId, name, price, stockQty, reorderLevel	checkStock(), updateStock(), isLowStock()
Cart	cartId, studentId, items[]	addItem(), removeItem(), checkout(), clear()
Order	orderId, studentId, status, totalAmount	create(), updateStatus(), cancel()
OrderItem	orderItemId, orderId, productId, qty, unitPrice	calculateSubtotal()
Admin	adminId, role	manageProducts(), processOrder(), generateReport()
EmailService	smtpConfig	sendConfirmation(), sendLowStockAlert()

4.5 Order State Diagram

Figure 5 — UML State Diagram showing all possible Order states and valid transitions:

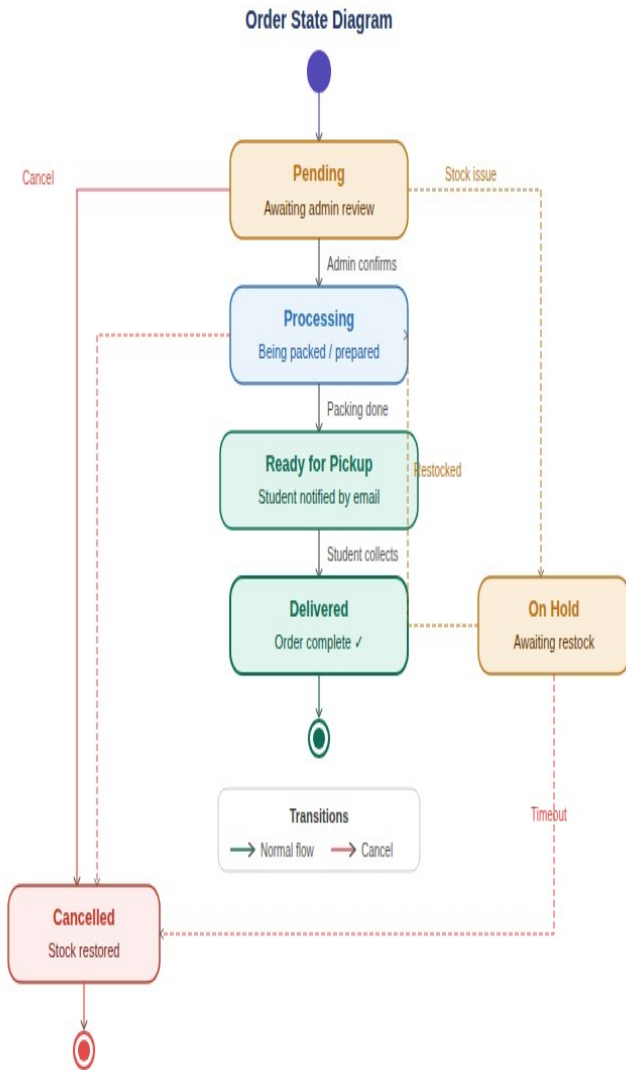


Figure 5: Order State Diagram — Lifecycle of an Order

4.6 Order Processing Flowchart

Figure 6 — Detailed flowchart showing the complete order processing flow with decision branches:

Order Processing Flowchart

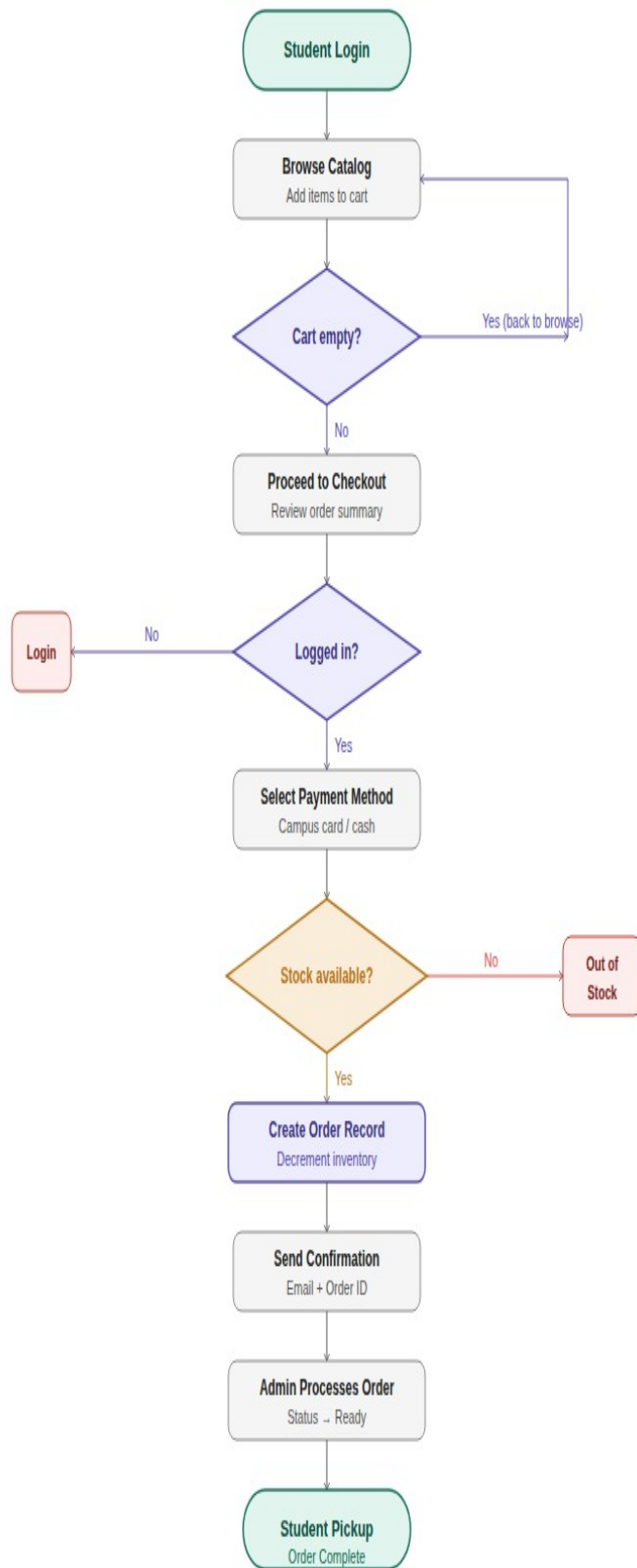


Figure 6: Order Processing Flowchart

4.7 Interface Design

Data Validation Mechanisms

Following Dennis et al. Ch.11 input validation principles, the following checks are implemented across all screens:

Check Type	Applied On	Validation Rule	Error Message Shown
Format Check	Registration — Email field	Must match university domain pattern: *@college.edu.eg	Invalid email format. Use your university email.
Completeness Check	Checkout — Payment & address	Payment method and pickup location must be selected before submit	Please complete all required fields.
Range Check	Cart — Quantity input field	Must be integer > 0 AND $\leq$ current stock_qty. Decimals rejected.	Quantity must be between 1 and available stock.
Existence Check	Login — Email field	Email must exist in Users table before password is checked	No account found with this email address.
Consistency Check	Checkout — Order total	Calculated total must match $\text{sum}(\text{qty} \times \text{price})$ server-side	Order total mismatch. Please refresh and retry.

Screen 1 — Student Catalog Page

Component	Specification
Navbar	Logo Search bar Cart badge Login/User menu
College Tabs	Engineering Medical — filters entire catalog
Sidebar	Category tree Price slider In-stock toggle
Product Grid	3 cols desktop / 2 tablet / 1 mobile. Card: image + name + price + stock + Add-to-Cart
Low-Stock Badge	Amber warning if stock $<$ reorder_level

Screen 2 — Admin Dashboard

Component	Specification
Metric Cards	Today's Orders Monthly Revenue Low-Stock Items Registered Students
Orders Table	Last 10 orders: ID Student Date Total Status Actions
Bar Chart	Daily order count — last 7 days

Component	Specification
Alert Panel	Red: products below reorder level with Restock button

4.8 Role-Based Access Control

Feature	Student	Admin	Faculty
Browse Catalog	✓	✓	✓
Shopping Cart	✓	✗	✗
Place Order	✓	✗	✗
Track Order	✓	✗	✗
Admin Dashboard	✗	✓	✗
Manage Products	✗	✓	✗
Process Orders	✗	✓	✗
Generate Reports	✗	✓	✓

Chapter 5 — Implementation Phase

Textbook Reference: Dennis et al. Ch.13 (Construction) & Ch.14 (Deployment)

5.1 REST API Endpoints

Method	Endpoint	Description	Auth
POST	/api/auth/register	Register new student account	No
POST	/api/auth/login	Authenticate → return JWT	No
GET	/api/products	List products (filter/sort/paginate)	No
GET	/api/products/:id	Get single product detail	No
POST	/api/cart/items	Add item to cart	Student
POST	/api/orders	Place new order (checkout)	Student
GET	/api/orders/:id	Get order + status	Student/Admin
PUT	/api/orders/:id/status	Update order status	Admin
POST	/api/admin/products	Create new product	Admin
PUT	/api/admin/products/:id	Update product	Admin
GET	/api/reports/sales	Generate sales report	Admin/Faculty

5.2 Module Breakdown

Module	Files	Responsibility
Auth	auth.js, login.html	JWT login, bcrypt hashing, session management
Catalog	catalog.js, catalog.html	Product listing, search, filter, sort, pagination
Cart	cart.js, cart.html	Cart management, stock validation, quantity controls
Orders	orders.js, checkout.html	Checkout, order creation, email trigger, tracking
Admin	admin.js, admin-*.html	Product CRUD, inventory panel, order processing
Reports	reports.js, reports.html	SQL-driven reports, chart.js, CSV export
Email	mailer.js	Nodemailer: confirmations, alerts, notifications

5.3 Test Plan

Test ID	Module	Test Case	Expected Result	Status
UT-01	Auth	Register valid college email	Account created	✓ Pass
UT-02	Auth	Login with wrong password	401 Invalid credentials	✓ Pass
UT-03	Catalog	Search 'oscilloscope'	EE-003 returned	✓ Pass
UT-04	Cart	Add qty > available stock	409 Insufficient stock	✓ Pass
UT-05	Orders	Place order → verify inv. decrement	Stock reduced in DB	✓ Pass
UT-06	Email	Order placed → email received	Email in < 30 s	✓ Pass
UT-07	Inventory	Stock drops below reorder level	Admin alert email fired	✓ Pass
UT-08	Reports	Generate monthly sales report	Renders in < 5 s	✓ Pass

5.4 Deployment Plan

Phase	Activity	Owner	Duration
Pre-Deploy	Code review, security audit, DB backup	Dev Team	2 days
DB Migration	Run CREATE/ALTER scripts, seed reference data	DBA	1 day
App Deploy	npm install --prod, configure .env, start PM2	DevOps	1 day
SSL + Nginx	Configure reverse proxy, install Let's Encrypt cert	DevOps	4 hrs
UAT	10 students + 2 admins test all key flows	Team+Users	3 days
Go-Live	Switch DNS, 24-hr monitoring, smoke tests	Dev Team	1 day

Chapter 6 — Product Catalog

6.1 Engineering College — Electronics & Communications

Category	Sample Products	For
Measurement	Digital Multimeter UT61E, Oscilloscope 2ch 100MHz, LCR Meter, Clamp Meter	Yr 2–4
Microcontrollers	Arduino Uno R3, Raspberry Pi 4, STM32 Discovery, ESP32 Dev Kit	Yr 3–4
Components	Resistor Kit E24 (600pcs), Capacitor Kit, Diode Assortment, Inductors	Yr 1–3
Power Supplies	DC Bench Supply 30V/5A, Variable Regulator Kit	Yr 2–4
Soldering	Soldering Station 60W, Solder Wire, Flux Paste, Desoldering Pump	Yr 2–4
RF & Comms	SDR Receiver Dongle, SMA Connector Kit, RF Signal Generator	Yr 4

6.2 Medical College

Category	Sample Products	For
Diagnostic	Stethoscope, Otoscope, Ophthalmoscope, Clinical Penlight	Yr 3–4
Lab Equipment	Microscope Slides 100pk, Cover Slips, Gram Staining Kit, Dissection Set	Yr 1–2
PPE	Lab Coat, Nitrile Gloves 100pk, Safety Goggles, Face Shield	All
Anatomy Models	Mini Skeleton, Heart Model, Brain Model, Eye Model	Yr 1–2
Vital Signs	Manual Sphygmomanometer, Clinical Thermometer, Pulse Oximeter	Yr 3–5

6.3 Sample Products with Pricing

SKU	Name	College	Price EGP	Stock	Status
EE-001	Digital Multimeter UT61E	Eng.	450	85	Active
EE-002	Arduino Uno R3	Eng.	280	120	Active
EE-003	Oscilloscope 2ch 100MHz	Eng.	3,200	3	⚠ Low Stock
EE-004	Breadboard 830-point	Eng.	45	300	Active

SKU	Name	College	Price EGP	Stock	Status
EE-005	Resistor Kit 600pcs	Eng.	120	200	Active
EE-006	Soldering Station 60W	Eng.	380	40	Active
MD-001	Stethoscope	Med.	550	60	Active
MD-002	Reflex Hammer Taylor	Med.	85	90	Active
MD-003	Dissection Set 8-piece	Med.	195	75	Active
MD-004	Lab Coat (per size)	Med.	150	100	Active
MD-005	Manual Sphygmomanometer	Med.	320	4	 Low Stock

Chapter 7 — Conclusions

7.1 SDLC Coverage Summary

Phase	Book Chapter	Deliverables
Planning	Ch.3	Feasibility (4 types), Charter, Sprint Plan, Gantt, Risk Register
Analysis	Ch.4	12 FRs, 7 NFRs, Interviews, Survey (n=120)
Analysis	Ch.5	14 Use Cases, UC-07 Spec, Sequence Diagram, State Diagram
Analysis	Ch.6	DFD L0 (Context), DFD L1 (5 processes), DFD L2 (P3 decomp.)
Design	Ch.9	3-Tier Architecture, REST API 17 endpoints
Design	Ch.10	ERD (8 entities, 7 relationships), Data Dictionary
Design	Ch.11	3 Wireframes, Navigation map, RBAC table
Implementation	Ch.13	SQL Schema, 7 Modules, Unit & System Test Plans
Deployment	Ch.14	7-phase Deployment Plan, UAT strategy
Agile	Scrum	Backlog (12 stories, 71 pts), 4 sprints, Burndown Chart

7.2 Future Enhancements

14. Native Mobile App (Flutter): Push notifications, barcode scanner for product lookup.
15. AI Recommendation Engine: Suggest tools based on enrolled courses and history.
16. Supplier API Integration: Auto-generate purchase orders at reorder level.
17. Student Rental Module: Short-term loans for expensive equipment (oscilloscopes).
18. Faculty Tool Lists: Instructor publishes required list; students add all to cart in one click.

7.3 References

- Dennis, A., Wixom, B. H., & Roth, R. M. (2015). Systems Analysis and Design, 6th ed. Wiley.
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- Nielsen, J. (1994). Usability Engineering. Morgan Kaufmann.

Appendix A — Interview Questions

For Students

19. What tools are required for your lab courses this semester?
20. Where do you currently purchase or borrow these tools?
21. How much time per semester do you spend sourcing tools?
22. What is your average monthly spending on academic equipment?
23. What features would be most important in a college online store?

Appendix B — Survey Results (n=120)

Question	Agree	Disagree	Neutral
Difficulty finding required tools?	87 (72.5%)	18 (15%)	15 (12.5%)
Would use an online college store?	110 (91.7%)	5 (4.2%)	5 (4.2%)
Tool cost is a barrier to performance?	95 (79.2%)	15 (12.5%)	10 (8.3%)
Prefer campus pickup over delivery?	89 (74.2%)	21 (17.5%)	10 (8.3%)
Tools unavailable when needed?	98 (81.7%)	14 (11.7%)	8 (6.7%)